25901155

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Elective (DSE) - MG4DSECHEA01 - CRIME SCENE INVESTIGATION, MANAGEMENT AND RECONSTRUCTION
(2024 ADMISSION ONWARDS)

Duration: 2 Hour **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **5** questions

Each question carries **4** marks

1. Mention one type of case where forensic evidence is crucial. [E] / [CO1]

Homicide case (murder):

- Helps identify the victim and suspect using DNA or fingerprints
- Determines cause and time of death
- Reconstructs the sequence of events (crime scene analysis)
- Provides strong scientific evidence in court

2. Give one example of documentary evidence in indoor scenes. [U] / [CO2]

Suicide note found inside a room:

- Provides written information about the incident
- May indicate motive or intention
- Helps in identifying handwriting or author
- Can be used as legal evidence in court

3. State four measures to prevent cross-contamination of evidence. [K] / [CO4]

Measures to prevent cross-contamination of evidence:

- Wear gloves, masks, and protective clothing at the scene
- Use separate tools for collecting different samples
- Package each piece of evidence in separate, labeled containers
- Avoid touching, sneezing, or talking directly over evidence

4. Illustrate a type of visual output from a 3D crime scene reconstruction that can be used in court to help jurors visualize the scene. [C] / [CO5]

Example: 3D animated reconstruction (crime scene walkthrough)

- Shows a virtual walkthrough of the crime scene from different angles
- Helps jurors clearly visualize layout and positions of objects
- Can demonstrate sequence of events step-by-step
- Makes complex evidence easier to understand in court

5. Mention the importance of portable evidence collection tools. [K] / [CO4]

Importance of portable evidence collection tools:

- Allow quick collection of evidence directly at the crime scene
- Help preserve evidence in its original condition
- Reduce risk of contamination or loss during transport
- Enable efficient and accurate documentation of evidence

6. What is a hypothesis? [E] / [CO1]

- A tentative or proposed explanation for an observation or problem
- Based on available facts or evidence
- Can be tested through experiments or investigation
- Helps guide further research or analysis

7. What is the first priority at any crime scene? [S] / [CO3]

First priority at any crime scene: Protection of life.

- Provide medical aid to injured persons
- Ensure safety of victims, officers, and public
- Remove any immediate danger (weapons, hazards)
- Call emergency services if required

[4x5 = 20]

Part B

Short Essay

Answer any **5** questions

Each question carries **7** marks

8. Describe the importance and functions of a Forensic Photographer at the crime scene. [K] / [CO4]

Importance and functions of a Forensic Photographer at a crime scene:

- Accurately documents the crime scene as found, without altering any evidence
- Captures overall, mid-range, and close-up photographs of the scene and evidence
- Provides a permanent visual record for investigation and court proceedings
- Helps in reconstructing the sequence of events later
- Photographs injuries, victims, and suspects for detailed analysis
- Uses proper lighting, angles, and scales to ensure clarity and accuracy
- Ensures photographs are properly labeled, stored, and maintained as legal evidence

9. Explain the similarities between the reconstruction process and the scientific method. [C] / [CO5]

Similarities between the reconstruction process and the scientific method:

- Both start with observation: collecting evidence or noticing phenomena.
- Both involve forming a hypothesis: proposing a possible explanation of events (crime scene or scientific phenomenon).
- Both require data collection: gathering physical evidence or experimental data systematically.
- Both use analysis: examining evidence or data to identify patterns and relationships.
- Both involve testing the hypothesis: reconstructing the scene or conducting experiments to see if the explanation holds.
- Both aim to reach a conclusion based on evidence rather than assumptions.
- Both emphasize documentation and reproducibility: recording procedures so findings can be reviewed or presented in court or scientific publication.

10. Describe logical approach to managing & testing several competing hypotheses. [C] / [CO5]

Logical approach to managing & testing several competing hypotheses:

- List all possible hypotheses based on the evidence and observations at the scene.
- Evaluate each hypothesis for consistency with known facts and available data.
- Prioritize hypotheses by plausibility and relevance to the case.
- Design tests or experiments to confirm or refute each hypothesis systematically.
- Collect and analyze evidence objectively to avoid bias toward any single hypothesis.
- Eliminate hypotheses that are contradicted by evidence, narrowing down to the most likely explanation.

- Document the process clearly to show reasoning, testing, and conclusions for investigators and the court.

11. What details should be included in evidence labels? [E] / [CO1]

Details to include in evidence labels:

- Case number – links the evidence to the specific investigation.
- Item number or description – identifies the type of evidence clearly.
- Date and time of collection – records when the evidence was secured.
- Name or initials of collector – shows who collected the evidence.
- Location of collection – specifies the exact place the evidence was found.
- Signature of the collector – ensures accountability and authenticity.
- Chain of custody information – tracks each person who handles the evidence to maintain integrity.

12. Explain briefly integration of documentation methods. [S] / [CO3]

Integration of documentation methods:

- Combines notes, sketches, photographs, and videos to create a complete record of the crime scene.
- Ensures consistency and cross-verification between different types of documentation.
- Helps capture details that may be missed by one method alone (e.g., a sketch shows layout, photos show exact appearance).
- Supports evidence preservation for analysis and court presentation.
- Facilitates clear communication among investigators, forensic experts, and legal personnel.
- Allows efficient reconstruction of the crime scene later using multiple reference points.
- Enhances credibility and reliability of the investigation report.

13. Explain the need for interstate coordination in criminal investigations. [K] / [CO4]

Need for interstate coordination in criminal investigations:

- Sharing information and intelligence between states helps track suspects who move across borders.
- Access to specialized resources and forensic laboratories not available in one state.
- Ensures consistent legal procedures and avoids jurisdictional conflicts.
- Facilitates joint operations for complex or organized crimes.
- Prevents duplication of efforts and saves time and resources.
- Enhances evidence collection and preservation by standardizing practices across states.
- Improves chances of successful prosecution by combining expertise and data from multiple jurisdictions.

14. Describe how forensic evidence from a vehicle can be used to connect a suspect car to a hit-and-run scene. [C] / [CO5]

Using forensic evidence from a vehicle to connect a suspect car to a hit-and-run scene:

- Tire marks analysis – comparing skid marks or tread patterns at the scene with the suspect vehicle's tires.
- Paint transfer – matching paint chips from the vehicle to damage on objects or victims at the scene.
- Glass fragments – linking broken vehicle glass found at the scene to the suspect car.
- DNA or biological evidence – detecting blood, hair, or skin cells from victims on the vehicle.

- Fibers or debris – connecting clothing fibers, soil, or debris on the car to the crime location.
- Vehicle parts or tools – identifying unique damage or mechanical features consistent with the scene.
- Digital evidence – using GPS, dashcam, or onboard computer data to track the vehicle's movements.

[7x5 = 35]

Part C

Essay

Answer any **1** questions

Each question carries **15** marks

15. Write a note on analysis of evidence in crime scene methodology. [S] / [CO3]

Analysis of Evidence in Crime Scene Methodology:

- Collection of Evidence – Gather all physical, biological, chemical, and digital evidence carefully.
- Preservation of Evidence – Use gloves, proper tools, and containers to prevent contamination or deterioration.
- Labeling and Documentation – Clearly label each item with case number, description, collector's name, and date/time.
- Maintaining Chain of Custody – Track everyone who handles the evidence to ensure authenticity in court.
- Classification of Evidence – Categorize evidence as physical, biological, chemical, trace, or digital for proper analysis.
- Preliminary Examination – Inspect evidence for visible characteristics such as fingerprints, blood stains, or damage.
- Forensic Laboratory Analysis – Conduct specialized tests: DNA profiling, fingerprint comparison, ballistics, chemical testing, and digital forensics.
- Comparison with Known Samples – Match evidence to suspects, victims, or objects from the crime scene (e.g., tire tracks, fibers).
- Interpretation of Results – Analyze findings objectively to establish relevance and connections to the crime.
- Integration with Documentation – Combine analysis with sketches, photographs, and videos for comprehensive scene reconstruction.
- Reconstruction of Events – Use evidence analysis to recreate the sequence and timing of events.
- Reporting – Prepare detailed, clear, and accurate reports for investigators and court proceedings.
- Legal Admissibility – Ensure that evidence and analysis meet legal standards for court use.
- Preservation for Future Use – Store evidence safely for potential re-examination or appeals.
- Importance – Supports suspect identification, strengthens prosecution, prevents miscarriages of justice, and ensures credibility of the investigation.

16. Explain ethical responsibilities of forensic scientists in evidence handling and reporting. [K] / [CO4]

Ethical Responsibilities of Forensic Scientists:

- Integrity and Honesty – Always provide truthful, unbiased findings.
- Objectivity – Base all conclusions solely on scientific evidence.
- Confidentiality – Keep case details and evidence private, sharing only with authorized personnel.
- Proper Evidence Handling – Collect, preserve, and transport evidence without contamination.
- Maintain Chain of Custody – Ensure evidence authenticity for court presentation.
- Competence – Work only within areas of expertise.
- Professionalism – Stay updated with current scientific methods and legal standards.
- Transparent Reporting – Clearly document methods, results, and limitations.
- Clarity for Legal Use – Make reports understandable to lawyers, judges, and jurors.
- Avoid Conflicts of Interest – Maintain independence from personal, financial, or external pressures.
- Legal Accountability – Follow laws, professional codes, and ethical guidelines.
- Court Testimony – Provide honest and accurate evidence in legal proceedings.
- Respect for Human Rights – Handle sensitive materials ethically and with dignity.
- Avoid Misleading Conclusions – Clearly state uncertainties or limitations in analysis.
- Maintain Public Trust – Ethical conduct reinforces credibility of forensic science and justice.

[15x1 = 15]